

Overview

This repository provides replication files for Cao, Lu, and Wu, "Synthetic Control Inference
→ for Staggered Adoption."

It is organized into two main components

- ****Simulation****: replication of the results in Section 3, including Figure 1.
- ****Intergovernmental coordination and criminality****: replication of the results in Section
→ 4, including Table 1 and Figures 2 and 3.

Data availability

All data sources are publicly available.

The crime and cartel data for Guanajuato used in Section 4 are provided by Alcocer (2024) and
→ are hosted on the Harvard Dataverse ([doi:10.7910/DVN/PBTDDM](https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi%3A10.7910%2FDVN%2FPBTDDM)).

Statement about Rights

We certify that the author(s) of the manuscript have legitimate access to and permission to
→ use the data used in this manuscript.

We certify that the author(s) have documented permission to redistribute and publish the data
→ contained in this replication package. The empirical data are publicly available from
→ Harvard Dataverse, and no restricted or confidential data are included.

Replication Instructions

Software and `gsynth` version note

The current verification of the revised Figure 3 was run on macOS 26.4.1 (arm64) with R 4.5.2
→ and MATLAB R2026a Update 2. RStudio is not required; the R scripts can be run from
→ RStudio or from an R console/terminal.

The submitted replication package was tested with `gsynth` version 1.4.0. The code may run
→ under other versions of `gsynth`, but exact numerical reproduction of the GSC estimates
→ in Figure 3 requires version 1.4.0. Older versions, including version 1.2.1, can produce
→ different GSC estimates.

Simulation Study

Access the folder `Simulation` to replicate this section. The folder includes the code needed
→ to reproduce Figure 1 that compares staggered synthetic control (SSC) and alternative
→ synthetic control variants through simulation studies.

Software requirements

- Current local environment used for this revision check: macOS 26.4.1 (arm64)
- R (4.5.2); RStudio is optional
 - required packages: R.matlab, doParallel, foreach, doRNG, MASS, nleqslv, dplyr,
→ parallel, gsynth, augsynth, ggplot2, patchwork
 - Use `gsynth` version 1.4.0 for exact reproduction of the submitted results; see the
→ software and `gsynth` version note above.
 - All required R packages can be installed by running command
→ `install.packages("package\_name")` The exception is augsynth, which should be
→ installed with `devtools::install\_github("ebenmichael/augsynth")`.

- MATLAB (R2026a Update 2)
 - required toolbox: Optimization Toolbox

Replication Steps

1. Generate Simulation Datasets:

- From the `Simulation` folder, run `sim\_sampling.R` in R or RStudio to create
 - ↳ simulation datasets.
- Output files will be saved in the `data/` directory.

2. Run Simulations:

- In MATLAB, run `sim\_scs.m` from the `Simulation` folder for SSC results. Then run
 - ↳ `sim\_alternative\_methods.R` from the `Simulation` folder in R or RStudio for
 - ↳ alternative methods.
- Output CSV files will be saved in the `output/` directory.

3. Plot Results:

- In R or RStudio, run `plot\_results.R` from the `Simulation` folder to generate
 - ↳ **Figure 1** from the simulation results.
- **Figure 1** will be saved in `output/` directory

Expected Running Times

The running times below were recorded in the original replication benchmark using MacBook Pro
 ↳ 2019, Intel Core i7, 16GB memory.

Program	Expected Running Time (seconds)
sim_sampling.R	53
sim_alternative_methods.R	5417
sim_scs.m	3795
plot_results.R	11

Detailed Directory Structure

- `sim\_sampling.R` - R script for sampling procedures in simulations.
- `sim\_scs.m` - MATLAB script for simulating staggered synthetic control.
- `sim\_alternative\_methods.R` - R script for simulating alternative methods.
- `plot\_results.R` - R script to generate plots from simulation results.
- `data/` - Contains simulation datasets generated by `sim\_sampling.R`:
 - `sim\_data.mat`, `sim\_data.RData` - Simulation data in MATLAB and R formats.
- `functions/` - Contains R and MATLAB functions used in simulations:
 - `att\_ci.m`, `att\_event\_ci.m` - MATLAB functions to implement SSC for overall ATT
 - ↳ effect and event-time ATT effect with inference
 - `ssc.m`, `att\_event.m` - MATLAB functions to implement SSC for individual treatment
 - ↳ effect event-time ATT effect without inference
 - `synthetic\_control.m`, `synthetic\_control\_batch.m` - MATLAB functions to estimate
 - ↳ synthetic control weights for one unit and a batch of units (i.e., all units)
 - `ssc\_inference.m` - MATLAB functions to implement SSC inference
 - `vline.m` and `vline\_license.txt` - MATLAB functions for plotting and related license
 - `sampling.R` - R functions to generate simulation sample
 - `padding\_matrix.R` - R functions for storing simulation results
 - `plot.R` - R functions for plotting
- `output/` - Contains simulation results
 - Estimation results as CSV files, named by simulation parameters and method (e.g.,
 - ↳ `sim\_results\_N33\_T15\_r3\_asy.csv`).
 - `Figure1\_simulation\_results.png` - Figure 1 that plots simulation results
 - `running\_time\_sim\_sampling.txt`, `running\_time\_sim\_alternative\_methods.txt`,
 - ↳ `running\_time\_sim\_scs.txt`, `running\_time\_plot\_results.txt` - Files that record
 - ↳ running time

Intergovernmental Coordination and Criminality

Access the folder `Intergovernmental coordination and criminality` to replicate this section.

- This section revisits the empirical analysis in Alcocer M (2025), "Increasing Intergovernmental Coordination to Fight Crime: Evidence from Mexico."

The folder includes the data and code needed to reproduce the main tables and figures, specifically:

- Table 1: Smallest eigenvalues of the sample analogue of the design matrix.
- Figure 2: Treatment scheme for different outcomes.
- Figure 3: Treatment effects estimated using staggered synthetic control (SSC) and generalized synthetic control (GSC) methods.

Software requirements

- Current local environment used for this revision check: macOS 26.4.1 (arm64)
- R (4.5.2); RStudio is optional
 - required packages: dplyr, gsynth, panelView, ggplot2, cowplot
 - The submitted replication package was tested with `gsynth` version 1.4.0. Older versions of `gsynth`, including version 1.2.1, may still run but can produce different GSC estimates in Figure 3.
- MATLAB (R2026a Update 2)
 - required toolbox: Optimization Toolbox

Replication Steps

1. Prepare cleaned data (R).
 - From the `Intergovernmental coordination and criminality` folder, run
 - `data\_cleaning.R` in R or RStudio.
 - This script loads the raw data from `raw\_data/`, applies the same sample restrictions as in the paper, and writes cleaned CSV files to
 - `smallest\_eigenvalues/cleaned\_data`, `treatment\_effect/cleaned\_data`, and
 - `treatment\_scheme/cleaned\_data`.
2. Compute smallest eigenvalues of the sample analogue of the design matrices (MATLAB).
 - Open directory `smallest\_eigenvalues/`.
 - In MATLAB, run `calculate\_eigenvalue\_for\_design\_matrix.m` from the
 - `smallest\_eigenvalues/` folder.
 - This script reproduces **Table 1**. It reads the cleaned data and summarize the smallest eigenvalues for each outcome.
 - **Table 1** will be saved in `smallest\_eigenvalues/output/` directory.
3. Visualize treatment scheme (R):
 - Open directory `treatment\_scheme/`.
 - From the `treatment\_scheme/` folder, run `plot\_treatment\_scheme.R` in R or RStudio.
 - This script reproduces **Figure 2**, which displays the timing of treatment adoption for homicide, theft, and cartel outcomes.
 - **Figure 2** will be saved in `treatment\_scheme/output/` directory.
4. Estimate treatment effects (R and MATLAB).
 - Open directory `treatment\_effect/`.
 - From the `treatment\_effect/` folder, run `estimation\_gsc.R` in R or RStudio.
 - The script computes the results by GSC method. It uses the "gsynth" package and
 - the cleaned data in `treatment\_effect/cleaned\_data` to compute event-time ATT
 - estimates and confidence intervals for all outcomes, and saves `results\_gsc.csv`.
 - In MATLAB, run `estimation\_ssc.m` from the `treatment\_effect/` folder.
 - The script computes the results by SSC method. It uses the SSC functions in
 - `treatment\_effect/functions` and the cleaned data in
 - `treatment\_effect/cleaned\_data` to compute event-time ATT estimates and
 - confidence intervals for all outcomes, and saves `results\_ssc.csv`.

- In MATLAB, run `plot\_results.m` from the `treatment\_effect/` folder.
 - This script reproduces **Figure 3** that compares the two methods: it reads
 - ↳ `results\_ssc.csv` and `results\_gsc.csv`, plots figures that compare SSC and GSC
 - ↳ event-time ATT estimates across all outcomes.
- Estimation results and **Figure 3** will be saved in `treatment\_effect/output/` directory.

Expected Running Times

The running times below were recorded in the original replication benchmark using MacBook Pro
 ↳ 2019, Intel Core i7, 16GB memory.

Program	Expected Running Time (seconds)
data_cleaning.R	2
calculate_eigenvalue_for_design_matrix.m	12
plot_treatment_scheme.R	4
estimation_gsc.R	315
estimation_ssc.m	10
plot_results.m	9

Detailed Directory Structure

- `raw\_data/` - Contains the raw data for all analyses.
 - `psrm\_cartel\_data.RData` - Annual cartel-related data, obtained from the Dataverse replication archive (doi:10.7910/DVN/PBTTDM).
 - ↳ `psrm\_cartel\_data\_raw.csv` - Non-proprietary CSV copy of `psrm\_cartel\_data.RData`.
 - `psrm\_crime\_data.RData` - Monthly crime data, obtained from the same Dataverse replication archive.
 - ↳ `psrm\_crime\_data\_raw.csv` - Non-proprietary CSV copy of `psrm\_crime\_data.RData`.
- `data\_cleaning.R` - R script that processes the raw data in `raw\_data/` and saves cleaned data to analysis folders.
 - ↳ `smallest\_eigenvalues/` - Contains data and code to compute the smallest eigenvalues of the sample analogue of the design matrices and export Table 1.
 - `cleaned\_data/` - Cleaned outcome data produced by `data\_cleaning.R`.
 - `psrm\_cartel\_data.csv` - Cleaned dataset for cartel outcomes.
 - `psrm\_crime\_data.csv` - Cleaned dataset for homicide and theft outcomes.
 - `calculate\_eigenvalue\_for\_design\_matrix.m` - MATLAB code that builds the sample analogue of the design matrices, computes their smallest eigenvalues, and writes the summary table.
 - ↳ `functions/` - MATLAB functions used for implementing SSC method.
 - `att\_ci.m`, `att\_event\_ci.m` - Implement SSC for overall ATT and event-time ATT effects with inference.
 - ↳ `ssc.m` - Implements SSC for individual treatment effects and event-time ATT without inference.
 - `synthetic\_control.m`, `synthetic\_control\_batch.m` - Estimate synthetic control weights for one unit and for a batch of units.
 - `ssc\_inference.m` - Implement SSC inference
 - `vline.m`, `vline\_license.txt` - Plotting utility and its license.
 - `output/` - Output folder for the eigenvalue calculations.

- ``Table1_smallest_eigenvalue.csv`` – Table 1: smallest eigenvalues of the sample
 - ↳ analogue of the design matrix for each outcome.
- ``running_time_calculate_eigenvalue_for_design_matrix.txt`` – Running time for
 - ↳ eigenvalue calculation.
- ``treatment_scheme/`` – Contains code and data that produce Figure 2, which visualizes the
 - ↳ timing of treatment adoption.
- ``cleaned_data/`` – Cleaned outcome data produced by ``data_cleaning.R``.
 - ``psrm_cartel_data.csv`` – Cleaned dataset for cartel outcomes.
 - ``psrm_crime_data.csv`` – Cleaned dataset for homicide and theft outcomes.
- ``plot_treatment_scheme.R`` – Plot the treatment scheme for different outcome groups
 - ↳ and save it as Figure 2.
- ``output/`` – Output folder for the treatment scheme figure.
 - ``Figure2_treatment_scheme.png`` – Figure that visualizes the treatment scheme for
 - ↳ different outcome groups.
 - ``running_time_plot_treatment_scheme.txt`` – Running time for treatment scheme
 - ↳ plotting.
- ``treatment_effect/`` – Contains data and code to estimate treatment effects and to plot
 - ↳ Figure 3.
- ``cleaned_data/`` – Cleaned outcome data produced by ``data_cleaning.R``.
 - ``psrm_cartel_data.csv`` – Cleaned dataset for cartel outcomes.
 - ``psrm_crime_data.csv`` – Cleaned dataset for homicide and theft outcomes.
- ``estimation_ssc.m`` – MATLAB script that estimates treatment effects by SSC method and
 - ↳ saves results.
- ``estimation_gsc.R`` – R script that estimates treatment effects by GSC method and
 - ↳ saves results.
- ``plot_results.m`` – MATLAB script that reads SSC and GSC results and produces Figure 3
 - ↳ that compares the methods.
- ``functions/`` – MATLAB functions used for implementing SSC method.
 - ``att_ci.m``, ``att_event_ci.m`` – Implement SSC for overall ATT and event-time ATT
 - ↳ effects with inference.
 - ``ssc.m`` – Implements SSC for individual treatment effects and event-time ATT
 - ↳ without inference.
 - ``synthetic_control.m``, ``synthetic_control_batch.m`` – Estimate synthetic control
 - ↳ weights for one unit and for a batch of units.
 - ``ssc_inference.m`` – Implement SSC inference
 - ``vline.m``, ``vline_license.txt`` – Plotting utility and its license.
- ``output/`` – Output folder for the SSC and GSC estimation.
 - ``results_gsc.csv`` – Estimated treatment effects by ``estimation_gsc.R`` and related
 - ↳ statistics from the GSC approach.
 - ``results_ssc.csv`` – Estimated treatment effects by ``estimation_ssc.m`` and related
 - ↳ statistics from the SSC approach.
 - ``Figure3_application_results.png`` – Figure that compares the SSC and GSC
 - ↳ approach, produced by ``plot_results.m``.
 - ``running_time_estimation_gsc.txt``, ``running_time_estimation_ssc.txt``,
 - ↳ ``running_time_plot_results.txt`` – Running time for GSC estimation, SSC estimation
 - ↳ and plotting.

Data citations

Alcocer, Marco. 2024. "Replication Data for: Increasing Intergovernmental Coordination to
→ Fight Crime: Evidence from Mexico." Harvard Dataverse. <https://doi.org/10.7910/DVN/PBTTDM>.

Contact

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